

SST MAKERSPACE COMPETITION BRIEF

V2



MAKERSPACE 3.0

OCTOBER 2026



SST
MAKERSPACE



SCHOOL OF
SCIENCE AND
TECHNOLOGY
PAN-ATLANTIC UNIVERSITY

Inspire



Famify

Optimize



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About SST Makerspace

SST Makerspace is a student-led innovation hub operating under the School of Science and Technology (SST) at Pan-Atlantic University. Our primary purpose is to provide hands-on opportunities for students to learn, design, build, and innovate practical solutions through competitions, workshops, and collaborative projects. We are deeply committed to inspiring creativity and confidence in engineering and science students, bridging the gap between theory and practice through real-world design projects, and promoting collaboration across disciplines. Ultimately, we strive to build a culture of innovation that aligns with both academic rigor and industrial needs.

Our Mission and Vision

At SST Makerspace, we empower students to transform theory into practice through immersive, hands-on learning and interdisciplinary collaboration. Our mission is to nurture a vibrant community of innovative thinkers and problem-solvers who develop sustainable, hardware-driven solutions that address real-world challenges and transform Africa's technological landscape. We envision a future where African ingenuity is at the forefront of engineering and hardware innovation. We aspire to be a pioneering hub that redefines the narrative of Africa as a land of builders, fostering an ecosystem of creative leaders who drive economic growth, societal progress, and resilient, locally engineered technologies.

The Challenge and Our Solution

SST Makerspace exists to bridge the gap between academic learning and real-world innovation by providing a structured community for hands-on research, prototyping, and product-driven experimentation. We create an environment where students, faculty, and industry partners collaborate to solve practical problems, develop engineering depth, and translate ideas into functional, market-relevant hardware solutions. By emphasizing applied learning, teamwork, and entrepreneurial thinking over grade-centric outcomes, the Makerspace cultivates technical competence, research discipline, and innovation readiness. Importantly, SST Makerspace is being intentionally developed as a scalable and replicable model, serving as a template for how universities can build sustainable, industry-aligned hubs that drive meaningful research, innovation, and hardware entrepreneurship.

Our Values and Community

Our community is driven by core values that prioritize learning by doing, entertainment, and team spirit. We believe in gaining skills relevant to future development while maintaining a focus on practical problem-solving. While our primary target audience includes engineering students, we welcome any university student in Nigeria interested in hardware, IoT, and robotics.

Future Prospects

Looking towards the future, SST Makerspace aims to become a launchpad for useful, locally relevant hardware products in Nigeria and beyond, where student-built solutions move beyond prototypes into real-world application. Our competitions are a strategic tool, designed not just to test performance, but to ensure learning remains fun, engaging, and deeply practical, while steadily building the skills required for serious hardware development. By combining structured challenges with the new Product Category, the Makerspace is positioning itself as a space where innovation is nurtured intentionally, talent is refined through experience, and viable hardware products can be developed, validated, and prepared for market impact.

Our Partners

Our initiatives are supported by a dedicated group of sponsors, including Total Energies, myAI, Auptex, Sages and Scribes, Pan-Atlantic University, Cedric Masters, Cors system, and Emmysoluk Electric Nig Ltd.

What to Expect from SST Makerspace 3.0

The SST Makerspace 3.0 delivers a transformative engineering experience that bridges the gap between classroom theory and real-world application, while also introducing participants to the journey of turning ideas into usable products. Students engage in the hands-on design and development of functional hardware systems, building strong competence in mechanics, electronics, embedded systems, and system integration, all within realistic constraints.

Beyond technical skills, participants develop professional and entrepreneurial capabilities through teamwork, leadership, communication, and problem-solving under pressure. With guidance from mentors, industry professionals, and business experts, especially within the Product Category, students learn how to refine designs, validate use cases, and think critically about feasibility, users, and value creation. The competition environment balances healthy rivalry with collaboration, ensuring learning, creativity, and innovation are rewarded alongside performance.

Participants can expect recognition, prizes, mentorship opportunities, and strong portfolio outcomes, particularly for teams whose projects demonstrate real-world relevance and market potential. Ultimately, students become part of a growing community of builders and innovators, gaining the confidence, network, and mindset needed to pursue advanced engineering challenges and contribute meaningfully to Nigeria's hardware innovation ecosystem.

SST Makerspace 3.0 Roadmap

SST Makerspace 2026 Event

Competition Category

The Rope Runner Challenge



15 teams compete.



Build a bot that moves along a rope.



Completes a set of predetermined tasks.



Teams are awarded across different categories.

Product Category

The Maker-to-Market Initiative



6 teams build products.



Addresses real world problems for SMEs.



Exhibition takes place on the event day to showcase the final products.



The teams with the most appraisal from attendees gets crowned best product.

Competition Category

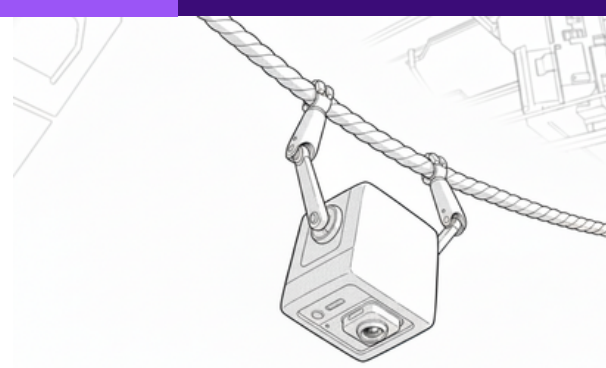
The Rope Runner Challenge





Compete
Among the Best!

Competition Overview



Synopsis

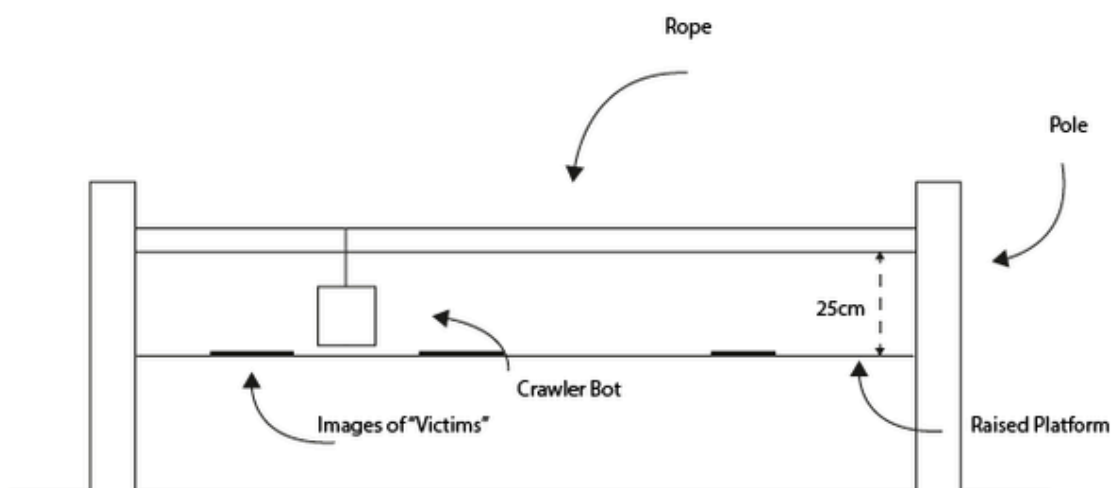
The Rope Runner Challenge is designed to inspire students to think creatively about how robots can move across suspended cables or ropes, a task that requires a careful balance of grip, stability, and efficiency. While the challenge may look simple at first glance, it reflects real engineering problems faced in industries today. Around the world, cable-based robotics are being developed to inspect power lines, maintain communication cables, and even deliver goods across difficult terrain. They are also used in rescue situations to transport supplies across rivers, ravines, or disaster zones where ground access is impossible.

For Nigeria, this competition is especially relevant. As the nation continues to expand its infrastructure and energy networks, reliable inspection and maintenance of power lines and transmission cables is critical to keeping the economy running. At the same time, disasters such as flooding, which affect many communities each year, highlight the need for innovative solutions to move goods and provide aid across blocked or unsafe areas. Encouraging young engineers to explore cable-traversing robotics creates a space to imagine practical, low-cost solutions that can be deployed in real Nigerian contexts.

Competition Format

The aim of this competition is to design and build an inverted walker robot capable of hanging from a rope and traversing it. The robot must maintain balance, overcome obstacles, and perform specific missions that mimic real-world challenges such as traversal stability and search-and-rescue operations.

The rope will be 8 meters long, suspended between two poles. Each team's robot must complete assigned missions while adhering to design rules and competition standards.



Side view of the track

Eligibility and Registration

Who Can Enter

Student Status: The competition is open to all active undergraduate students (or students pursuing an equivalent or similar degree) of universities/colleges in **Africa**. Participants must be currently enrolled at the time of registration.

Technical Background: There are no strict major restrictions; however, teams are expected to have the capability to design, build, and program a robot that meets the mechanical and electrical specifications..

Team Composition

Team Size: Each team must consist of at least 3 and at most 5 members. Single-member teams are not permitted to ensure collaboration and safety.

Roles: Teams are encouraged to distribute roles among members (e.g., Mechanical Lead, Electrical/Software Lead, Project Manager, etc.), though this is not strictly enforced.

Substitutions: Once registered, team members can only be substituted with written approval from the competition organizers.

- No substitutions are allowed within 48 hours of the competition start date.
- A team member cannot belong to more than one team simultaneously.

Registration Process

To officially enter the Rope Runner Competition, teams must complete the following steps:

- **Form a Team:** Gather 3-5 members and agree on a Team Name.
- **Complete the Online Form:** Fill out the Team Registration Form.
- **Application Link:** [Apply here](#)
- **Submission:** Submit the form before the deadline (*Incomplete registrations will not be processed*).

Registration Deadlines & Penalties

- Registration Opens: 20th January, 2026.
- Registration Closes: 15th April, 2026. (at 11:59 PM).

Late Submission Policy:

Late registrations may be accepted up to 2 days after the indicated registration deadline but will incur a penalty of -5 points from the final mock score.

No registrations will be accepted after the "Late Submission" window closes to allow for logistics planning.

Eligibility and Registration



Fees, Refunds & Withdrawal Policy

Registration Fee: A sum of ₦5,000.00 per team will be required to finalize your commitment to competing.

Team Name and “SST Makerspace Competition 2026” should be included in the description during payment.

Account Details

Account Name: SST Makerspace

Bank: Ecobank

Account Number: 0010170420

Material Costs: Teams are responsible for sourcing their own materials (motors, sensors, PLA). Teams are also responsible for arranging their logistics to and fro the event venue.

Venue: SST [School of Science and Technology] Foyer, Pan-Atlantic University, Ibeju-Lekki , Lagos.

Withdrawal: If a team wishes to withdraw, they must notify the organizers via email by at least 4 weeks before the main competition day. There is possibility for refunds to be processed if legitimate grounds are stated i.e logistics issue to the venue.

Confirmation & What Teams Receive

Upon successful registration, the Team Lead will receive a Confirmation Email containing:

- Official Team ID: (e.g., Team-01) – *required for labeling the robot and other process stated later in the document.*
- *Other details will be sent to teams after mock qualification.*



The Rope Runner Challenge is our **affordable, locally relevant alternative** to drone solutions for inspection and monitoring.

– Joseph Adigwe, Technical and Documentation Lead, SST Makerspace



Selection Process (Qualification)

While registration is open to all interested teams, the main competition day is a high-stakes showcase limited to the **top 15 qualifying teams**. Registering does not automatically guarantee a spot in the final race; teams must prove their technical readiness through a rigorous vetting process.

- **Video Submission:** Teams must submit a clear, high-resolution video of their robot in motion. The video must unequivocally demonstrate the robot hanging from and traversing a rope segment using the correct mechanism (no wheels) and other technical requirements. *See here for more details*
- **Selection Criteria:** The top 15 spots will be awarded to the teams that demonstrate the most reliable locomotion, strict adherence to safety/design rules, and robust build quality during the screening phase.

Note: Only teams that pass the qualification stage (Video Review) will be cleared to compete for the grand prizes on the main event day.

Awards

Recognition in the SST Makerspace competition is designed to celebrate not just the fastest robot, but the most well-rounded engineering solution. Prizes will be awarded across several categories to honor speed, intelligence, and design quality.

OVERALL WINNER

Awarded to the teams with the highest cumulative score from both the main competition's activities (1st, 2nd and 3rd places).

ENGINEERING DESIGN AWARD

A technical prize for the team demonstrating the most robust chassis construction, efficient drive mechanism, and superior cable management.

Important Dates & Milestones

S/N	Activity	Date
1	Design Session, Panel & Competition Launch	6 th December, 2025
2	Competition Registration - Form Opens	12 th January, 2026
3	Competition Initial Clarification and Virtual Coffee I	18 th February, 2026
4	Virtual Coffee II Sages & Scribes Consultants Session I	25 th February, 2026
6	Sages & Scribes Consultants Session II	18 th April, 2026
5	Competition Registration - Form Closes	15 th April, 2026
7	Q&A Session II (Competition Focus) & Virtual Coffee III	6 th June, 2026
8	Mock Event (Video Submission Deadline)	17 th August, 2026
9	*Main Event (D-Day)	5 th – 9 th October, 2026

Deadlines & How to Request Clarifications or Extensions

- An e-mail has to be sent to the organizing committee (The SST Makerspace Executive Team) about the particular request and the reason for it.
- The SST Makerspace Executive Team reserves all right to either permit or dismiss requests, and extensions will be no more than a week from the day the team receives and acknowledges the request.

For enquiries, email: sstmakerspace@pau.edu.ng

Follow updates and reminders on our Instagram: [@sstmakerspace](https://www.instagram.com/sstmakerspace)

***Dates are tentative and will be communicated in due time.**

Activity One - Swiftplay

The Goal

The primary objective of Activity One is a test of pure mechanical efficiency and stability. Teams are challenged to navigate their crawler bot across the full length of the 8-meter competition rope from Point A to Point B. While the task involves reaching the finish line as quickly as possible, the environment is designed to simulate the unpredictability of real-world infrastructure. Robots must endure intentional disturbances, including tension variations, and controlled vibrations. This activity serves as the ultimate stress test for the robustness of the crawler mechanism, exposing any weaknesses in grip strategy, chassis balance, and power management.

Race Procedure

To begin a run, the robot must be positioned precisely behind the starting line. The run is officially initiated by the sound of a buzzer, at which point the timing begins. The bot must proceed continuously along the rope without external assistance. The run is considered complete the moment any part of the robot's chassis or body crosses the vertical plane of the finish line.

Checkpoints and Recovery Protocol

To balance difficulty with competitive fairness, fixed checkpoints are established at predetermined intervals along the rope. These serve as "save points" to prevent a single error from ending a team's run entirely.

The Fall Rule: If a robot loses its grip and slips off the rope, the run is not reset to zero. Instead, the team is permitted to manually retrieve the bot and reattach it at the last checkpoint successfully passed.

Time Penalty: The clock does not stop during this recovery process. Therefore, a fall naturally incurs a time penalty, as the team loses valuable seconds retrieving and resetting the bot.

Timing System and Transparency

To ensure absolute fairness and accuracy, the timing process is strictly governed by a dual-verification system:

Live Stopwatch: A large, visible digital stopwatch will run live during the attempt, allowing the audience and teams to track progress in real-time.

Hand Stopwatch: A designated official timekeeper will simultaneously operate a high-precision handheld stopwatch to cross-check the live time. The final recorded time will be verified against both devices. In the event of a close call or dispute, a video camera recording at 60fps or higher will serve as the official reference, ensuring that every team has equal assurance of accuracy regarding start and stop times.

Manual Intervention and Boundaries

Strict rules govern when a team member touches the robot.

- **Permitted:** Physical contact is allowed only when a fall occurs, and strictly for the purpose of repositioning the bot to the nearest passed checkpoint.
- **Prohibited:** Any manipulation intended to assist the robot such as pushing, adjusting its orientation, or stabilizing it during a wobble is strictly forbidden and will result in penalties or disqualification for that run.
- **Boundaries:** The robot must rely solely on the competition rope; gripping external supports or harnesses outside the rope boundaries is a violation of the rules.

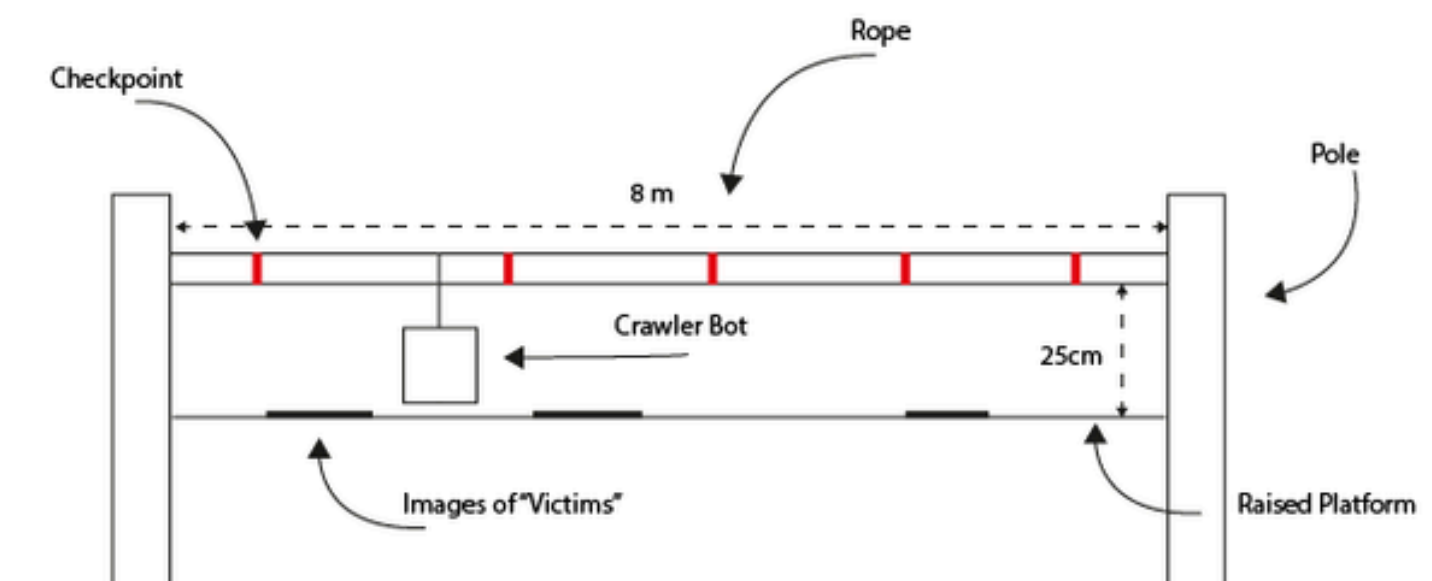
Scoring Breakdown

Scoring for Swiftplay is prioritized by speed, but rewards stability.

- **Speed:** The robot with the fastest traversal time secures the highest points, followed by the second and third fastest times.
- **Clean Run Bonus:** To reward engineering excellence, additional points are awarded for a "Clean Run", completing the track without a single fall or restart. Conversely, frequent falls that require manual resets will result in a lower placement due to the time lost.

Documentation

Judges will maintain a detailed run log for every attempt. This log records the official start time, finish time, total number of falls/resets, and any specific penalties applied. This ensures that the final leaderboard is based on a transparent, data-backed assessment of performance.



Activity Two - Search and Rescue

This activity simulates a post-disaster search-and-rescue mission where each crawler bot must navigate along the same rope used in Activity One but now perform the additional task of identifying “victims” positioned beneath the rope and transmitting the necessary evidence to a local server. The robot must use its bottom-facing camera to visually detect victims, capture a clear image, calculate the distance travelled from the start of the rope using wheel encoders, and then send both the image and its associated metadata to the competition server.

The Goal

The objective is to locate and correctly report as many victims as possible in the least time,

Victim Markers (size, color, placement rules)

Each victim is represented by a printed marker showing an injured person. To ensure consistent recognition across all robots, each marker will be printed on a clear white paper at a size between 12 and 15 cm on each side, using images with a resolution of 700–900 px (*See Appendix for Sample Images*). These markers will be placed flat on the floor beneath the rope, spaced randomly, not rotated significantly, and always fully visible, mixed with images of other “healthy” persons. This is to test model training integrity.

For the bot to capture them effectively, the camera should be mounted facing directly downward, maintaining a distance of roughly 22–28 cm from the ground. Images should be captured in JPEG format, ideally in Quarter Video Graphics Array (QVGA) resolution (320×240) for quick transmission, though VGA (640×480) may be used if a team is confident in the WiFi performance.

As the robot moves along the rope, its wheel encoders must keep an accurate log of the distance travelled. Competitors will compute this themselves by using the standard relationship between encoder ticks, wheel circumference, and forward displacement.

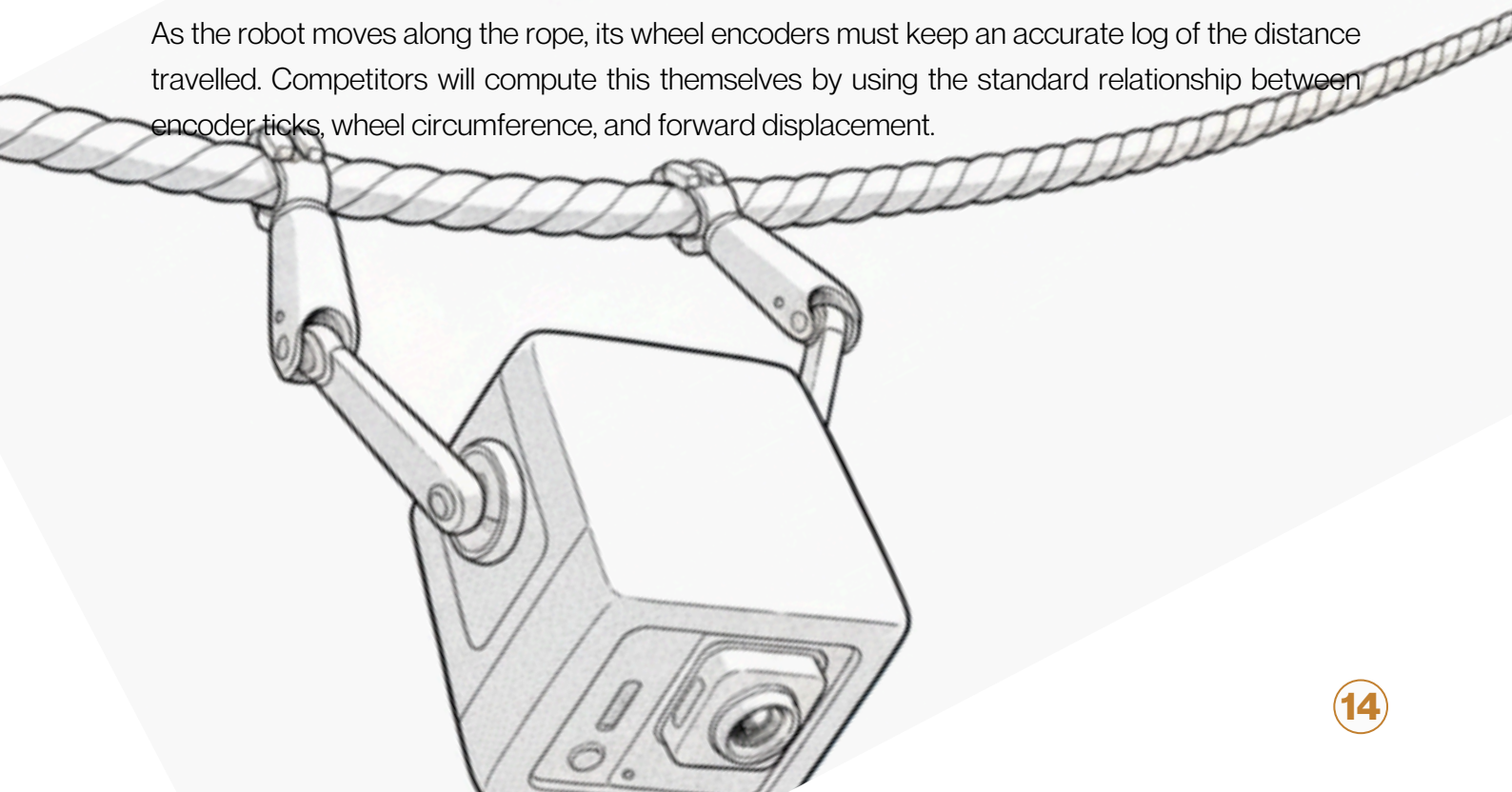


Image & Metadata Specification (file format, naming, mandatory fields, team tag overlay)

At the moment a victim appears in the frame, the robot must capture a photograph that clearly shows the victim and is free from blur or obstruction. The image must also include an overlaid team tag, written in a readable white font with a black outline and placed neatly in one corner; this tag must match the metadata the robot sends. Once an image is captured, the robot must package it in base64 format together with its team name, timestamp, and measured distance (in centimeters from the start of the rope) and transmit everything via HTTP POST to the local server's upload endpoint. A retry system should be implemented: if the upload fails, the robot attempts again several times with a short delay. The server will respond with a confirmation that the image and data have been received and accepted. Only images that display the victim clearly, include the proper team tag, are readable as valid JPEGs, and have plausible distance values will be counted.

Data Structure:

```
"team_id": "Phoenix",  
"victim_tag": "victim",  
"distance_cm": x,  
"timestamp": "2025-03-11T14:25:30Z",  
"image_id": "Phoenix_142530_234cm.jpg",  
"image_base64": "<base64>"
```

Other Info

To succeed in this activity, each team must ensure that their robot can reliably capture images, correctly overlay the team tag, measure distance accurately, and communicate with the server without corruption. A valid entry requires that the victim fills enough of the frame to be recognisable, the team information matches across both the image and metadata, and the robot maintains logical forward progression in its recorded distances. Lighting may vary slightly across the venue, so teams should avoid relying on auto-exposure alone and consider adding soft downward LEDs for stability. Image compression should be balanced: too high causes blur, too low results in long upload times. Teams should thoroughly test their JSON formatting, and encoder calibration before competition day.

Server details will be communicated to qualified teams after the mock activity.

Design Rules & Constraints

Power & Electronics

- Battery-operated only
- No external power cables, tethered supplies, or wired connections during competition.
- All power systems must be self-contained inside the robot.
- Allowed battery types include Li-ion, LiPo, LiFePO₄, NiMH.
- Not allowed; Lead-acid batteries, exposed battery terminals, damaged cells.
- Voltage & current safety - Max battery voltage recommended: 12–14.8 V (to avoid high-current hazards).
- All wiring must be fully insulated and securely routed.

Size & Weight

- Starting size limit: The robot must fit within 25 cm × 25 cm × 25 cm before deployment.
- Weight considerations - No fixed maximum weight, BUT:
 - The robot must not cause noticeable rope sagging.
 - The total load must not deform, fray, or overstress the rope.
- Size of the bot will be measured using a measuring cube and weight judged visually by rope deflection during inspection.

Mechanical Design Requirements

- Secure assembly
- All screws, parts, and structural components must be firmly attached.
- No dangling or loosely connected mechanisms.
- Permitted materials
 - Plastics (PLA, PETG, ABS), aluminium, steel, acrylic, wood, 3D printed parts.
- Prohibited materials
 - Sharp metal edges, sandpaper-like surfaces, fire-producing components, corrosive chemicals.
- Mechanisms must grip, clamp, or crawl on the rope.
- Wheels cannot roll along the top of the rope like a normal vehicle.
- The robot must HANG from the rope and not ride on top.

Autonomy & Control

- Full autonomy not required
- Remote control (Bluetooth, RF, Wi-Fi) is allowed during operation.
- All logic must be on-board (Microcontroller, sensors, encoders).
- Permitted actions
 - Pre-loaded code.
 - Autonomous adjustments guided by sensors.

Repairs, Modifications & Inspection

- On-site repairs allowed - before and after an activity
 - Teams may fix broken components between runs.
- You may bring
 - Extra batteries, spare motors, replacement sensors, small tools.
- Not allowed
 - Swapping out an entirely different robot mid-event.
 - Making major design changes after inspection.
- Any major component replacement must be approved by judges.



Rope & Course Specifications

Rope Characteristics

- Rope type
 - Synthetic braided rope (nylon or polyester)
- Diameter
 - Between 12–16 mm
- Length
 - Total rope length: 8 meters
- Distance from ground - 25cm

Mounting & Tension

The Rope is:

- Firmly tensioned between two strong poles.
- Fixed at a height ensuring clear visibility.
- Slightly inclined or horizontal depending on event setup.
- Important constraint:
 - The bot should NOT noticeably deform or sag the rope when hung.

Course Layout

Checkpoints

- Placed at fixed distances (see track layout image).
- Used for restart locations if a bot falls.

Obstruction

- Vibrations
 - Introduced at specific moments to simulate real-world instability.
 - All teams experience the same vibration pattern.

Environment

Lighting

- Consistent, controlled indoor lighting.

Safety

- Teams must remain behind designated lines while bots run.

General Rules

Robot Operation

- The robot begins at the designated start point and must traverse to the end., controlled wirelessly by a driver.
- No damaging or abrading the rope.
- If the robot falls, it must be manually returned to the last checkpoint reached by a race official.

Runs & Attempts

Each team receives:

- One official run per activity.
- One practice run before the event.

Retry conditions

Only permitted for:

- Battery failure
- Motor breakdown
- Unexpected malfunction

*** This is at the judges' discretion**

Retry time limit: 5 minutes to reset and restart.

Conduct & Fairness

- Judges' decisions are final as regards competition scoring and penalty allocations.
- Teams must show sportsmanship and must not tamper with the course, rope, or other teams' robots.
- Any unsafe behaviour leads to immediate disqualification.
 - i.e. unruly behaviour towards officiating personnel, or dissent towards judges or officials

Disallowed Behaviours

- Physically touching the robot mid-run.
- Intentionally shaking the rope or creating unfair advantages.
- Using materials that could damage the rope.

Evaluation Rubric

Scoring System

Category - Design (25%)

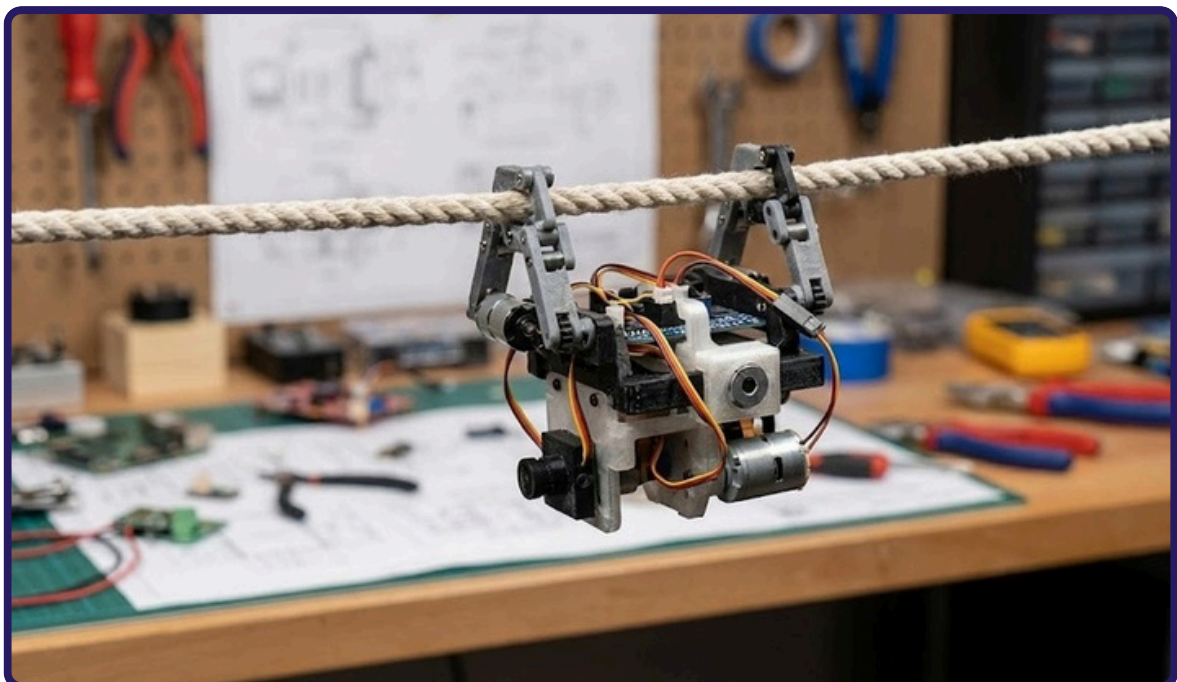
Sub-category	Max Points
Chassis Design	20
Drive Mechanism	20
Aesthetics & Presentation	20
Innovation / Extra Features	20
Obstacle Navigation	10
Electronics & Wiring	10
Design Subtotal	100

Category - Swiftplay (35%)

Sub-category	Max Points
Traversal Time (ranking-based)	100
Obstacle Handling & Stability	25
Completion Status	15
Swiftplay Subtotal	140

Category - Search & Rescue (40%)

Sub-category	Max Points
Number of Victims Detected (aggregate)	80
Accuracy & Tagging Quality (aggregate)	40
Reliability of Transmission (aggregate)	20
Speed Bonus (mission completion time)	20
Search & Rescue Subtotal	160
TOTAL	400



Bonus Points, Caps & Fairness Rules

- **Sustainability Bonus:** +6 points (applied to Design total) for demonstrable recycled/locally-sourced materials used as a clear design choice. Cap at 6 points.
- **No-Fall Bonus (Swiftplay):** +10 points if the robot completes traversal without any falls/restarts. (Included in the Swiftplay Completion Status allocation.)
- **First-Attempt (SAR Bonus):** +5 points per victim (within SAR scoring) if the image was transmitted and accepted on the first upload attempt. Cap this at +15 points across all victims.
- **Maximum Bonus Cap:** A team may receive up to +20 bonus points total (sum of all bonuses) ; beyond this, additional bonuses are not counted.
- **Fairness Rule:** All teams run under identical course conditions (rope tension, obstacle placement, lighting). Any deviation is recorded and compensated uniformly.

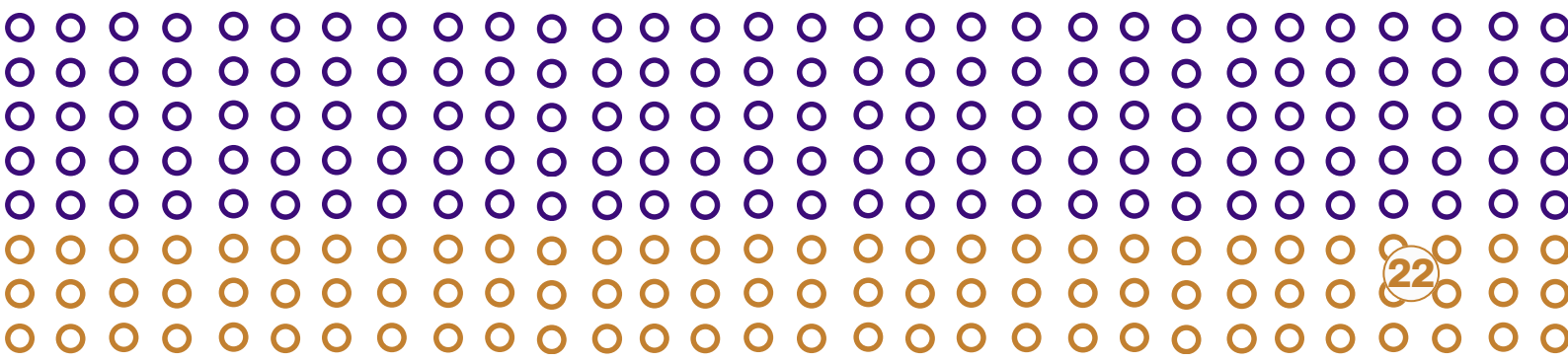
Tie-Breakers

If two or more teams have equal total scores, apply the following in order:

- Higher SAR subtotal (team with better search & rescue performance wins).
- Higher Swiftplay Traversal Time ranking (i.e., faster traversal).
- Higher Innovation/Extra Features score (Design subcategory).
- Fewest falls across Swiftplay (more stable bot wins).

Dispute Resolution & Appeals Process

- Teams must lodge appeals immediately after the posting of official results at the Judges' table.
- Appeals must state the reason and provide evidence if available (camera log, local video clip).
- Judges convene within 15 minutes to review recorded video and server logs.
- If necessary, a replay may be scheduled; replays are limited to one per appeal.
- The judges' panel issues a written, timestamped response within 30 minutes of appeal filing.
- All judge decisions are final; appeals can request reconsideration only for procedural errors, not for subjective judgment calls.



Penalties

General Penalties

Violation	Penalty
Unsafe design (exposed wiring, overheating battery, loose parts)	-30 points
Minor rule violation (non-critical spec breach)	-15 points
Major rule violation (size, rope damage, prohibited parts)	-50 points
External assistance during run	-40 points
Unsportsmanlike conduct (minor)	-20 points
Failure to fix safety issue after warning	Disqualification
Severe misconduct or disruption	Disqualification

Penalties

Activity 1 — Swiftplay Penalties

Violation	Penalty
Manual intervention while on rope	+20 seconds
Excessive swinging / rope twisting	-10 points
Did not reach endpoint	Traversal score = 0

Activity 2 — Search & Rescue (SAR) Penalties

Violation	Penalty
Blurry / unusable image	Image rejected (0 points)
Victim partially visible	-5 points per image
Incorrect victim label	-10 points per image
Missing label	-10 points per image
Missing team name/ID	Image rejected
Manual image upload / screenshot	Submission invalidated
Incorrect data format	-10 points

Mock Selection Procedure

Before the final competition lineup is announced, every team must pass through a mock selection phase designed to verify that the robot is not only well-conceived on paper, but also physically capable of performing the core task. The submission for this stage is a clear, high-resolution video **(1920X1080p)** showing the robot in motion. The video must unequivocally demonstrate that the bot is hanging from the rope and traversing it using the correct crawling or grip-and-pull mechanism, and that it does not use wheels to drive along the rope length. In addition, the robot must be shown meeting the key build rules, including proper rope safety, battery-only operation, and correct start configuration dimensions. The guide for this screening phase makes it clear that any failure in the compliance check can result in immediate disqualification or a requirement to re-engineer before final selection, and that the top spots go to teams that demonstrate the most reliable locomotion, strict adherence to safety/design rules, and robust build quality during screening.

Video Submission

The video must be clear, stable, and high-resolution. It should show:

- the robot at rest,
- the robot being measured,
- the robot being attached to the rope,
- the robot powered on,
- the robot moving forward on the rope,
- the robot's gripping mechanism in operation,
- a short overview of the software logic,
- the robot capturing an image and identifying it,
- the robot sending the image over a local network (to a locally hosted server),
- and at least a short traversal segment that proves the motion is real and controlled.

Technical Specification Sheet

Teams must also submit a short but exact technical summary of the robot, covering the mechanical, electrical, and software design. This is necessary so judges can compare what is shown in the video with what the team claims their robot is built to do.

Mechanical Section

This should clearly state:

- chassis size at start configuration,
- total bot weight,
- rope-gripping method,
- material choice,
- number of limbs/arms/contact points,
- rope diameter compatibility,
- how the bot avoids slipping,
- how it handles knots or rope vibration,
- and whether the structure is rigid enough to hold the electronics securely.

Electrical Section

This should include:

- battery type,
- battery voltage,
- battery capacity,
- power distribution method,
- motor type,
- motor driver used,
- current protection or safety measures,
- wiring method,
- and how power is delivered to the motors, controller, and sensors.

Software Section

This should explain:

- the control logic used for movement,
- whether the bot is autonomous or semi-autonomous,
- how encoder data is read,
- how the bot decides when to move or stabilize,
- any sensor use,
- and how the system handles timing, retries, or image/data transmission if your bot uses SAR-related features.

Beyond motion, judges also inspect whether the robot is likely to survive the main competition.

They check:

- chassis rigidity,
- whether all parts are properly secured,
- wire management,
- and the presence of any useful innovations such as stabilizers, smart tensioning, or other helpful mechanisms.

Recommended Submission Format

To keep the process consistent, each team should submit:

- a mock selection video,
- a technical specification sheet,
- a short build summary,
- and any supporting photos or diagrams.

A clean file naming format should be used, such as:

TeamName_MockSelection_Video.mp4

TeamName_TechnicalSheet.pdf

TeamName_BuildPhotos.zip

Deliverable Name	Content Requirements	Date	Focus
Review Stage			
Registration Deadline / Kickoff		April 15	Team confirmation & project start
Review 1	- Final concept sketch or CAD - Chassis concept - Material selection - Initial component list - Early prototype photos - Initial control or sensor plan	April 27	Concept validation & initial design
Review 2	Video Submission Required Resolution: 1920 X 1080p Duration : 3 - 5 minutes *Pledge Form	May 15	The video submission must include all members of the team talking about the bot building process and their expectations for the event. This helps the planning team put faces to the team members and publicity for the teams. The SST Makerspace encourages all teams to capture behind the scenes of their building process and include it in their video submission. This helps to preserve the memory of building and validates the process for the teams.
Review 3	- Partially or fully assembled bot - Motor and driver testing - Sensor wiring or encoder testing - Basic movement or climbing tests - First version of code or control logic - Updated BOM and budget - Problems discovered during testing	June 17	Structural build & component integration

Deliverable Name	Content Requirements	Date	Focus
Review Stage			
Review 4	- Fully assembled or nearly completed bot - Successful rope traversal tests - Stability and grip test results - Distance or timing test results - Camera/image capture setup if SAR is involved - Software/debugging status - Final list of unresolved issues	July 8	Functional subsystem testing Performance optimization with final tuning & pre-competition readiness
Final Submission / Readiness Check	- Mock Submission - Video Submission - Complete System Validation	August 17	Complete system validation

Next 3-Week Plan

Each report must end with a short plan for the next stage of development, showing:

- What will be built next
- What will be tested next
- What the team hopes to achieve before the next review

SST Makerspace 3.0 Bot Donation Pledge Form

As part of our effort to create meaningful social impact through SST Makerspace 3.0, and in collaboration with partner organizations dedicated to expanding access to technology for persons with disabilities and the less privileged, we invite participating teams to voluntarily donate their competition bots at the conclusion of the event.

This initiative is aimed at extending the value of your work beyond the competition by enabling your robot to be used for training, demonstrations, and skill development programs that empower others to gain hands-on experience in engineering and technology.

By signing this pledge, your team expresses its willingness to contribute to this cause and support the broader mission of using innovation as a tool for inclusion, education, and community development.

Pledge Statement

We, the undersigned team, hereby voluntarily agree to donate our competition robot developed for SST Makerspace 3.0 to support community impact initiatives facilitated by SST Makerspace and its partner organizations.

We understand that:

- This donation is entirely voluntary.
- There is no financial compensation guaranteed for this contribution.
- The donated robot may be used for educational, training, and demonstration purposes.
- Our team will receive formal recognition for our contribution to this initiative.

We make this commitment in the spirit of giving back, promoting inclusivity, and supporting the development of technical skills within underserved communities.

Authorization

Team Name: _____

Team ID: _____

Team Leader's Name: _____

Contact Email / Phone: _____

Team Leader Signature: _____

Date: _____

Thank you for contributing to a future where technology creates opportunities for all.

Technical Build Guidelines

System Architecture Overview - Mechanical, Electrical, Software

The Rope Runner bot operates as a unified, feedback-driven system where mechanical, electrical, and software subsystems function in a tight loop to navigate the 8-meter suspended track. The architecture begins with the mechanical subsystem, designed as an inverted walker that hangs from the rope to maintain a low center of gravity for passive stability. The chassis, fabricated from lightweight materials like 3D-printed PLA, is engineered to stay within a 30 cm cubed volume and support a 1kg payload while keeping the total weight under 2.5kg. Unlike traditional rolling robots, this system relies on specific "crawler arms" driven by DC motors to perform a grip-and-pull motion, enabling it to traverse the rope without utilizing wheels for linear movement.

Powering this physical structure is the electrical subsystem, which serves as the robot's nervous system. A rechargeable LiPo battery, typically 7.4V, provides the necessary energy, splitting into two distinct power rails: a high-current path feeding an L298N H-bridge driver to actuate the motors, and a regulated logic rail powering the microcontroller, such as an ESP32 or Raspberry Pi. This central processing unit serves as the brain, integrating sensory inputs from a bottom-facing camera and a wheel encoder to create a digital understanding of the environment below the rope.

The software subsystem acts as the orchestrator, running concurrent logic threads to manage both locomotion and mission-critical objectives. During the "Swiftplay" phase, the code focuses on motor synchronization and speed control via PWM signals to navigate obstructions like knots and vibrations. In the "Search and Rescue" phase, the software shifts to an intelligent surveillance role, fusing visual data from the camera with location coordinates from the encoder to identify, tag, and transmit images of "victims" to a local server using Wi-Fi. Seamless system integration is achieved when the software processes these sensory inputs in real-time to adjust mechanical torque and speed, ensuring the bot remains stable and responsive while executing complex data transmission tasks without pausing its journey.

Note that these guidelines are just a way to help you get started with building your bot and are not the exact way every bot must be built

Mechanical Design Guidance (chassis, grip types, materials, stiffness)

To get this inverted walking thing to work reliably, we're using a clever mechanical setup we call the "double-sided inverse shaft connected to a central rotating shaft." The whole point of this is to keep things simple by using a single motor and avoiding the headache of trying to perfectly sync up multiple motors.

How the Drive System Works:

The core idea is a single primary shaft, driven by one motor (we'll figure out the motor specs later), that just spins in one direction at a constant speed. We attach special components to the ends of this central shaft. These components are designed to hold two other shafts in a weird, inverse orientation, one sticking up (positive) and one sticking down (negative). This inverse geometry is the key! It's what turns the simple spin of the central shaft into the complex, alternating leg movements we need for the robot to walk upside down.

Materials and Making It:

We strongly recommend using 3D printing (Additive Manufacturing) for those critical, inversely-oriented parts. The materials should be either PLA or ABS plastic.

- Why PLA/ABS? These plastics are a great choice because they give us the perfect mix of high precision (accurate prints), easy manufacturing, and enough strength and structural integrity to handle the operational stress.
- Precision is Key: We need a super-high-quality 3D print. If there's any mistake or misalignment in those external shaft geometries, the robot's "walk" will be off, and the whole system will perform poorly.

Structure and Integrity:

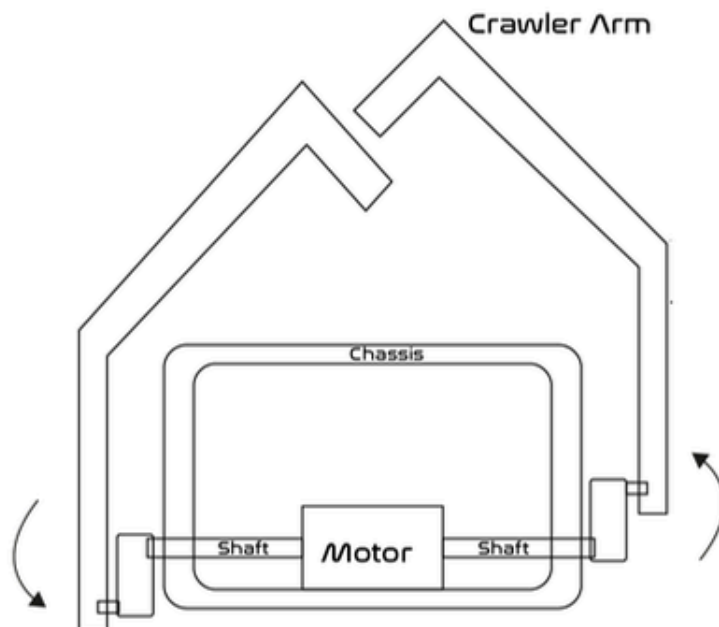
The overall structure is a huge factor in how long and how well the robot performs.

- The Main Body: The chassis needs to be big and tough enough to hold and protect all the electrical bits—motor control, microcontroller, battery, and wiring. Everything must be securely mounted so nothing shakes loose when the robot is moving.
- The Legs and Rope Grip: The robot's legs need specific features to grab the climbing rope effectively. We have to build in a mechanism or design allowance that ensures a strong, reliable grip on the rope. This is essential for maximizing friction and preventing slippage, which would stop the robot's climb. Good examples would be textured pads, simple clamping features, or a curve that perfectly fits the rope's diameter.

Keeping It Stable:

A major design goal is making sure the robot is stable, both when standing still and when walking. We need to prevent any unnecessary tipping or swaying that could make it lose its grip or tangle up the mechanism.

- Center of Gravity (CG) Control: The design team needs to carefully analyze where all the weight is placed to keep the robot's Center of Gravity as low and central as possible.
- Extra Stability: If just placing the components doesn't give us the stability we need, we might have to add a secondary measure. A simple, passive "hanger" or tail system at the back of the robot can act as a counterbalance. Just remember, if we add a tail or any stabilizing element, it counts as a full, functional part of the robot and must be included in the final size and weight calculations for the competition rules.



Sample Technical Diagram of the bot

Designing the electrical system for your crawler bot involves creating a compact, reliable, vibration-resistant power and control architecture that can operate autonomously throughout both competition activities. Since the robot runs entirely on batteries, the electronics must deliver stable power to motors, sensors, and the control unit while maintaining safe temperatures, clean wiring, and predictable behaviour under load.

Power System & Battery Integration

Your battery selection must supply enough continuous current for the motors and electronics without voltage drops during high-load moments such as climbing over knots or compensating for rope vibrations. A LiPo or Li-ion pack sized between 7.4V–12V is ideal, depending on your motor system. The battery should feed into a regulated bus: a high-current line for motors and a clean 5V/3.3V line for control electronics. Voltage regulators must be physically separated from high-current motor wiring to avoid noise injection. All connectors should lock securely, and the battery must be mounted inside an insulated, vibration-proof compartment with strain relief on all leads.

Motor Driving & Control Electronics

The robot must use a motor driver capable of handling the stall current of its motors. Drivers such as TB6612FNG, Cytron 10–13A units, or any H-bridge with appropriate thermal performance are preferred. Mount the driver close to the motors to minimize losses, and keep logic lines shielded or twisted to reduce interference. For teams using the ESP32, ensure the PWM pins are mapped correctly, and isolate motor noise using capacitors across the motor terminals or a small LC filter if needed. The control board—Arduino, ESP32, or Raspberry Pi—must be securely mounted, away from sources of heat or mechanical stress, with accessible USB/UART ports for debugging.

Sensors, Encoders & Data Integration

For Activity 2, wheel encoders are required to determine the distance along the rope. Encoders must be rigidly aligned with the drive shafts; even small misalignments cause pulse skipping or jitter. Use hardware interrupts on your microcontroller to ensure accurate pulse counting at high speed. Distance is computed from $\text{pulse count} \times \text{wheel circumference} \times \text{gear ratio}$. This distance must be transmitted alongside the image data for victim documentation. The ESP32-CAM or external camera module must be oriented downward with consistent lighting; mount the camera at a fixed height (~6–10 cm above the target plane) to ensure consistent image size and focus. A short ribbon cable or directly mounted ESP32-CAM board is recommended to avoid movement blur.

Electrical Architecture & Communication

The robot should maintain two internal power lines: a high-current motor line and a clean logic line. Keep these electrically isolated except at the battery source. Use common grounds, but route heavy motor grounds separately to prevent noise feedback into the microcontroller. For SAR image transmission, the ESP32 or Raspberry Pi should form an HTTP/MQTT client sending bundled JSON + image payloads to the local server. The onboard architecture must handle task scheduling: motor control loop, encoder counting, and camera capture must not block each other. On ESP32, dual-core allocation or timed tasks (e.g., FreeRTOS) are ideal. On Arduino, offload camera to a separate MCU.

Wiring, Safety & Reliability

All wires on a crawler bot must withstand movement, shaking, and repeated impacts from obstacles. Use stranded silicone wires, not rigid ones. Keep wiring harnesses short and grouped using heat-shrink or cable sleeves. Any connection prone to disconnection (Dupont jumper wires) should be avoided or glued in place. Solder joints must be reinforced, and connectors should have mechanical strain relief. Place the main switch in an accessible location and include a simple fuse or polyfuse if running higher current systems. Before competing, every team should perform a vibration test: hang the robot on a rope, manually shake the rope, and confirm that no electronics reset or power drops occur.

Recommended Control Architectures

A dual-processor setup is ideal: an ESP32 or Arduino handling motion + encoders, and an ESP32-CAM handling imaging and network uploads. For advanced teams, a Raspberry Pi Zero 2 W can unify everything but must be thermally managed and given a stable 5V supply. The final architecture must ensure deterministic timing for motor control and data reporting.

Goal:

Reliably capture a clear image of a “victim”, pair it with an accurate distance reading, transmit both to the server, and have the server (authoritatively) detect and score the submission.

Camera placement & capture (ESP32-CAM)

- Mounting: rigidly mount the ESP32-CAM directly beneath the robot, pointing straight down. Eliminate wobble with a short bracket + rubber damping.
- Height: 22–28 cm above target plane (optimum). Verify with a test print so the victim fills ~25–40% of frame.
- Resolution/format: default QVGA (320×240) JPEG for speed; VGA (640×480) only if Wi-Fi is rock-solid. Aim for JPEG size <150 KB (quality ~60–75).
- Lighting: add a small downward LED ring to stabilize exposure

Image pre-processing & metadata

- Pre-send checks: basic brightness/blur check; crop to ROI if possible to reduce size.
- Mandatory metadata (with every image):
 - team_id (string)
 - distance_cm (numeric, from encoders)
 - capture_timestamp (ISO8601)
 - image_id (unique per image)
 - image (binary JPEG or base64)

Atomicity: send image + metadata in one POST; never separate them and expect server to match later.

Encoders(distance)

- Encoder: use a dedicated friction pulley driven encoder; measure and record `ticks_per_rev` and derive `distance_cm = (ticks/tpr)*circumference`.
- Mounting: rigid, spring-loaded idler to ensure consistent motor shaft contact.

Sync: capture timestamp and the current encoder tick at the instant of exposure — these two values must travel together.

Communication protocol & reliability

- Transport: HTTP POST to `/upload_detection` (multipart/form-data preferred) on the local server. Include X-API-Key: `<teamkey>`.
- Retry policy: exponential backoff up to 5 attempts (e.g., 0.5s, 1s, 2s, 4s, 8s). On persistent failure, queue to local storage (SPIFFS/SD) for bulk upload after run.
- Limits: keep images <150 KB and avoid flooding (rate-limit client to 1–2 uploads/sec).

Time sync: ESP32 should NTP-sync at boot; server time is authoritative for disputes. Always include device `capture_timestamp` and server will add `receive_timestamp`.

Model & training essentials

- Task: single class “victim” (keep classes minimal).
- Data collection: use the same camera and mount as competition. Collect 300–500 labeled images minimum (per class). Include 20–30% negative frames (no victim).
- Annotation: tight, consistent bounding boxes; use Labelling / CVAT / Roboflow.
- Split: 70% train / 20% val / 10% test; avoid session leakage (don’t split near-duplicate frames across sets).
- Augmentation: moderate brightness, slight rotation ($\pm 10^\circ$), motion blur, scale. Avoid unrealistic transforms.
- Model choices: lightweight detectors (YOLOv5n/YOLOv8n/MobileNet-SSD). Input size 320×320 or 640×640 depending on server capacity.
- Training: monitor precision, recall, mAP; favor recall (missed victims worse than false positives).
- Thresholding: choose confidence ~0.4–0.6; require minimum bbox area to reject noise.
- Validation: test under motion — run full robot captures, upload to server, verify detection rates in real conditions.

Wheel Encoder (distance measurement)

Assumptions

- Quadrature or single-channel encoder
- Encoder mounted on a friction pulley
- Known TICKS_PER_REV and pulley circumference

```
volatile long encoderTicks = 0;

#define ENCODER_PIN 34
#define TICKS_PER_REV 600
#define PULLEY_CIRCUMFERENCE_CM 12.56 // example: 4cm diameter
void IRAM_ATTR encoderISR() {
    encoderTicks++;
}

float getDistanceCM() {
    noInterrupts();
    long ticks = encoderTicks;
    interrupts();
    return (ticks / (float)TICKS_PER_REV) * PULLEY_CIRCUMFERENCE_CM;
}

void setupEncoder() {
    pinMode(ENCODER_PIN, INPUT_PULLUP);
    attachInterrupt(digitalPinToInterrupt(ENCODER_PIN), encoderISR,
    RISING);
}
```

Atomic image + metadata upload (HTTP POST)

Key idea:

Image and metadata must be sent together in one request.

```
#include <WiFi.h>
#include <HTTPClient.h>

#define SERVER_URL "http://192.168.4.1:5000/upload_detection"
#define API_KEY "TEAM_API_KEY"

bool uploadDetection(camera_fb_t* fb, float distanceCM, String imageID) {
    HTTPClient http;
    http.begin(SERVER_URL);
    http.addHeader("X-API-Key", API_KEY);

    String boundary = "----ESP32BOUNDARY";
    http.addHeader("Content-Type", "multipart/form-data; boundary=" + boundary);

    String payload =
        "--" + boundary + "\r\n"
        "Content-Disposition: form-data; name=\"team_id\"\r\n\r\n"
        "SST-RR-05\r\n"
        "--" + boundary + "\r\n"
        "Content-Disposition: form-data; name=\"distance_cm\"\r\n\r\n"
        + String(distanceCM, 2) + "\r\n"
        "--" + boundary + "\r\n"
        "Content-Disposition: form-data; name=\"image_id\"\r\n\r\n"
        + imageID + "\r\n"
        "--" + boundary + "\r\n"
        "Content-Disposition: form-data; name=\"image\"; filename=\"img.jpg\"\r\n"
        "Content-Type: image/jpeg\r\n\r\n";

    String closing = "\r\n--" + boundary + "--\r\n";

    int contentLength = payload.length() + fb->len + closing.length();
    http.addHeader("Content-Length", String(contentLength));

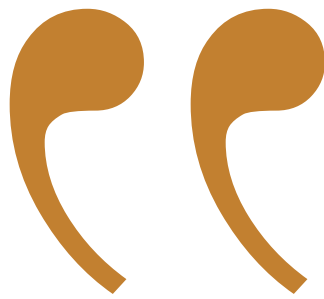
    WiFiClient* stream = http.getStreamPtr();
    http.POST(""); // initiate connection

    stream->print(payload);
    stream->write(fb->buf, fb->len);
    stream->print(closing);

    int code = http.GET();
    http.end();

    return (code == 200);
}
```

- **Chassis:**
 - Lightweight and durable material - 3D-printed PLA or selected material.
 - Crawler arms.
 - Compartment for electrical components.
- **Locomotion:**
 - Motors: 1 or 2 DC geared motors (e.g., 6V, 100-200 RPM, higher RPM for faster speeds) for dual motor action.
- **Control System:**
 - Microcontroller: Arduino Uno, ESP32, or Raspberry Pi.
 - Motor Driver: L298N or similar H-bridge motor driver.
 - Power Supply: Rechargeable LiPo battery (e.g., 7.4V, 1000mAh)
- **Sensors:**
 - Camera Module (e.g., ESP32-CAM, Raspberry Pi Camera): For FPV or image processing of Images.
 - Wheel Encoder for location sensing
- **Connectivity:**
 - Bluetooth Module (HC-05/06): For smartphone control.
 - Wi-Fi Module (ESP32 built-in): For web-based control or data streaming.
- **Software:**
 - Arduino IDE or VS Code with PlatformIO for programming.
 - Python for Raspberry Pi applications.



**Participating in the SST
Makerspace 2.0 Gripperbot
Competition allowed me to tap
into my handy and creative side,
a part of me that isn't usually
explored in the classroom. It was
a comprehensive and
unforgettable learning
experience.**



**– Michael Atourah, 500 Level
Electrical and Electronics
Engineering**

Frequently Asked Questions

Q: Can students from other universities participate in this category?

A: Yes. Unlike the Product Category, the Rope Runner Challenge is open to all active students of universities/colleges in Nigeria. You do not need to be a PAU student to compete in this category.

Q: Can we use wheels to drive along the rope?

A: No. The rules explicitly state that the robot must be an "inverted walker" that hangs from the rope. Wheels cannot roll along the top of the rope like a vehicle; the mechanism must "grip, clamp, or crawl."

Q: Is there a maximum weight limit?

A: There is no fixed numeric weight limit, but there is a performance limit. The robot must not cause noticeable rope sagging or deform/fray the rope. If your robot is too heavy and sags the line significantly during inspection, it may be disqualified.

Q: What if our battery dies or a motor breaks during a run?

A: You may be granted a 5-minute retry window to fix the issue and restart, but this is at the judges' discretion and is reserved for genuine mechanical/electrical failures, not code errors.

Q: What happens if our robot falls off the rope?

A: You are allowed to pick it up. The "Fall Rule" states that you can manually retrieve the bot and reattach it at the last checkpoint successfully passed.

Important: The clock does not stop during this recovery. A fall essentially acts as a time penalty because you lose seconds resetting.

Q: How do we qualify for the main event?

A: Registration alone does not guarantee a spot. You must submit a Video Entry demonstrating your robot successfully hanging from and traversing a rope. Only the top 10 teams from this video screening will be invited to the final competition.

Appendix

A - Documents

[Judge Scoring Sheet](#)

[Judge Scoring Rubric](#)

[Mock Race - Evaluation Rubric](#)

[Sample Images - Victims](#)

B - Resources

[Building a Crawler Bot](#)

[Inspiration Video for Building Bots](#)



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